

Trip Report PDF Generator - Project Specification

Project Goal

Create a simple desktop application for a truck driver to generate completed Trip Report PDFs from a pre-existing company PDF template.

The target user is not highly technical, so the application must prioritize simplicity and ease of use.

The user should never need to manually edit PDFs or create text boxes.

The application's sole purpose is:

1. Enter trip information.
2. Click Generate PDF.
3. Receive a completed Trip Report PDF.

No login system, database, cloud storage, or web server is required.

Technology Stack

Language

Java 21

UI Framework

JavaFX

PDF Library

Apache PDFBox

Build Tool

Maven

Project Structure

src/main/java/

```
app/  
    Main.java  
  
model/  
    TripEntry.java  
    TripReport.java  
  
ui/  
    MainView.java  
  
pdf/  
    PdfGenerator.java
```

resources/

```
templates/  
    TripReportForm2026.pdf
```

PDF Template

Use the provided PDF template:

Trip Report Form 2026.pdf

Important:

The following fields are already pre-filled in the template and must NOT be modified:

- Driver Name
- Driver Number
- Power Unit

The application only fills the remaining fields.

Always generate a new PDF file.

Never overwrite the original template.

User Interface Requirements

Create a single-window desktop application.

Window title:

Trip Report Generator

Layout:

Trip Information

Mileage Start: []

Mileage Finish: []

Total Mileage: [auto calculated]

Trip Start Date: []

Trip End Date: []

Trip Entries

[Table]

[Add Row]

[Remove Selected Row]

[Preview]

[Generate PDF]

Header Fields

Mileage Start

Numeric field.

Mileage Finish

Numeric field.

Total Mileage

Read-only.

Automatically calculated:

Total Mileage = Mileage Finish - Mileage Start

Trip Start Date

Date picker.

Trip End Date

Date picker.

Trip Entry Table

Create a JavaFX TableView.

Columns:

1. Mileage
 2. City / Zip Code
 3. Shipper
 4. Consignee
 5. Type
 6. Trailer Number
 7. Duty Code
-

Column Definitions

Mileage

Text field.

Numeric only preferred.

City / Zip Code

Text field.

Shipper

Text field.

Consignee

Text field.

Type

Dropdown (ComboBox)

Allowed values:

- EM
- TL
- LTL

These correspond to the checkboxes on the PDF.

Only one option may be selected.

Trailer Number

Text field.

Duty Code

Dropdown (ComboBox)

Allowed values:

- A
- B
- C
- D
- E
- F
- G
- H
- J
- K
- N
- S
- T

Row Limits

The PDF contains a fixed number of trip rows.

Determine the exact row count from the template.

The application should not allow more rows than fit on the PDF.

Disable Add Row when the maximum is reached.

Generate PDF Button

When clicked:

1. Validate user input.
2. Load PDF template.
3. Write values into correct locations.
4. Save generated PDF.

Output filename:

TripReport_YYYY-MM-DD_HH-mm.pdf

Example:

TripReport_2026-06-14_18-30.pdf

Save to:

User's Downloads folder

or

Prompt user for save location.

PDF Mapping Requirements

The implementation should use PDFBox.

Create a dedicated PdfGenerator class.

All PDF coordinates should be stored as constants.

Example:

HEADER_MILEAGE_START_X HEADER_MILEAGE_START_Y

HEADER_MILEAGE_FINISH_X HEADER_MILEAGE_FINISH_Y

ROW1_MILEAGE_X ROW1_MILEAGE_Y

etc.

Avoid magic numbers scattered throughout the code.

Checkbox Handling

The PDF contains checkboxes:

EM TL LTL

The UI uses a dropdown.

During generation:

If Type == EM

Place an "X" in EM checkbox position.

If Type == TL

Place an "X" in TL checkbox position.

If Type == LTL

Place an "X" in LTL checkbox position.

Only one checkbox should be marked.

Preview Feature

Preview button should:

1. Generate temporary PDF.
2. Open using system PDF viewer.

Purpose:

Allow the user to verify placement before saving final copy.

Validation Rules

Mileage Start and Mileage Finish:

- Required
- Numeric

Trip Start Date:

- Required

Trip End Date:

- Required

For each populated row:

- Mileage required
- City required
- Shipper required
- Consignee required
- Type required

- Duty Code required

Show user-friendly error messages.

Do not crash.

UI Design Goals

The application should NOT attempt to visually replicate the PDF.

It should behave like a clean data-entry form.

Requirements:

- Large readable fonts
- Clear labels
- Minimal clutter
- Simple workflow
- Suitable for non-technical users

The user experience should be:

Open app

↓

Enter trip data

↓

Generate PDF

↓

Done

Nice-To-Have Features (Optional)

Not required for Version 1.

Possible future enhancements:

- Remember last save location
- Remember window size
- Editable generated PDFs
- Auto-open generated PDF after save
- Keyboard shortcuts
- Dark mode

Do NOT implement databases, accounts, authentication, cloud storage, APIs, or networking.

Keep Version 1 focused on PDF generation only.

Deliverables

1. Complete Maven project.
2. JavaFX desktop application.
3. PDFBox integration.
4. Clean object-oriented design.
5. Comments explaining PDF coordinate mapping.
6. README explaining build and run steps.

Primary goal:

A truck driver should be able to generate a completed Trip Report PDF in under 2 minutes without touching PDF editing software.